



Circling Guidance (Optional)

Inset Runway Edge Light (LED)

High Intensity, Bi-directional, Inset Light

VAS-IREL-12-X/X-L-X-X

Compliance with standards

- ICAO Annex 14, Volume 1
- FAA AC 150/5345-46
- FAA Engineering Brief 67D
- IEC 61827

Application

- Lights installed on both sides of the runway to indicate the position of the runway edge. Use at night or in low visibility conditions.
- High-intensity, Bi-directional, Runway Edge lights that can be used for category I, II & III systems for Runways width of 45M as well as of 60M.

Specifications

- | | |
|-------------------------|------------------------------------|
| • Power Supply | 2.8A~6.6A |
| • Rated Power | W/W 42W, W/Y 44W, R/W 25W, R/Y 27W |
| • Power Factor | >95% at nominal current |
| • Temperature | -40° C to +55° C |
| • Life Time of LED | 50,000 hrs |
| • Protection Class | IP 67 |
| • Protrusion Height | 6.3mm |
| • Isolation Transformer | 45W |
| • Color | Red / White / Yellow |
| • Direction | Bi-directional |
| • Light fitting Size | Ø12" |

Features

- The light fitting is designed in strict accordance with the technical requirements and specifications of ICAO, IEC and FAA for Runway and Taxiway as Navigational aid light for Airports.
- A surge protection device is provided in the electronics as required by FAA "Engineering Brief 67D.
- The long life, energy saving and maintenance-free characteristics of LEDs bring economic benefits.
- High-efficiency power factor correction circuit ensures effective reduction of interference to the power supply network.
- Efficient design of LED heat dissipation effectively reduces the working temperature of light fitting, LED and drivers to ensure long service life.
- Fully dimmable lights, simulates the dimming curve as per Standard's requirement.
- Electrolytic capacitors of high temperature and long life are used in circuitry to improve the stability of the LED driver.
- The optical prism adopts special tempered glass to prevent the prism from bursting due to drastic changes in temperature.
- Fail-open option is available to achieve compatibility with advance control and monitoring systems. The LED light automatically disconnects from the secondary side of the isolation transformer, resulting in an open circuit condition.



Features

- The overall protection level of the light fitting reaches IP 67, preventing dust and rain from eroding the interior of the light.
- Fully compatible with existing airfield lighting series circuits.
- Color emitted directly by LEDs in absence of colored filters ensures no energy losses and no color shifts.
- The main body of the light fitting is made of aluminium alloy which is strong and has good anti corrosion performance.
- The height of light fitting protruding from the ground is only 6.3mm. This helps in strongly reducing vibrations to aircrafts and to light itself, increasing its lifespan.
- The upper cover of the light fitting adopts high strength design and drop forging process, with good mechanical properties, strong bearing capacity and impact resistance
- Fasteners are made of stainless steel which are durable.
- The parts are modularly designed and universal, which are easy to replace.
- Style L-823 connectors are used.

Ordering information

VAS-IREL-12-X/X-L-X-X

X = O/M

O = Without Monitoring

M = With Monitoring

X = 1/2

1= Without CG

2= With CG

L = Led Light

X/X = W/W, W/Y, R/W, R/Y

12 = 12" Diameter

IREL= Inset Runway Edge Light

VAS = VARDHMAN

Photometric Data

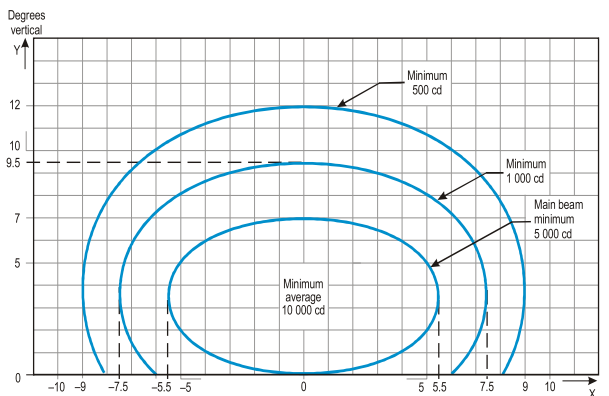


Figure A2-9. Isocandela diagram for runway edge light where width of runway is 45 m (White light)

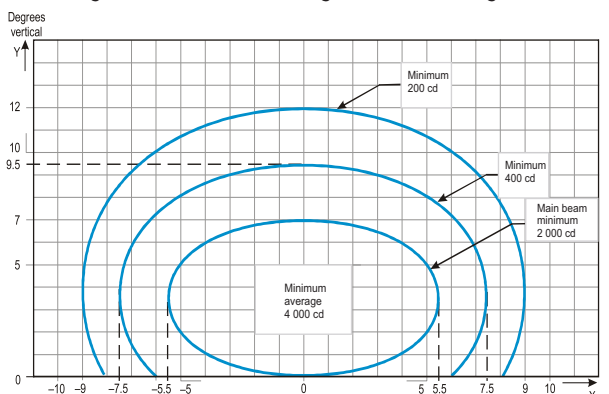


Figure A2-9. Isocandela diagram for runway edge light where width of runway is 45 m (Yellow light)

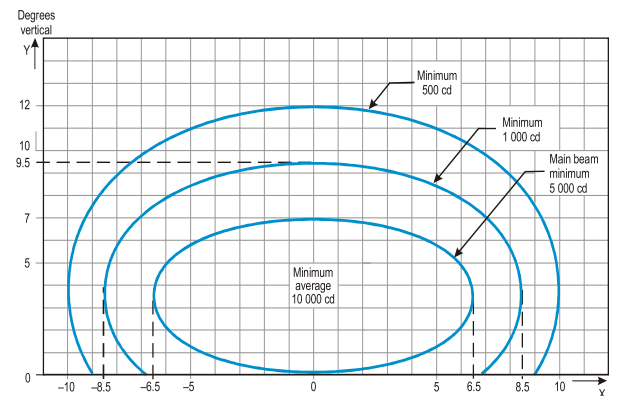


Figure A2-10. Isocandela diagram for runway edge light where width of runway is 60 m (White light)

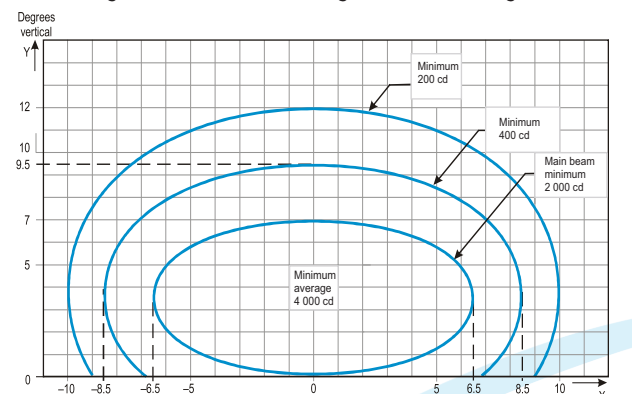


Figure A2-10. Isocandela diagram for runway edge light where width of runway is 60 m (Yellow light)



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Ver. 1.0

Rev. 1.1/Sep-2025 Product specifications and pictures may slightly vary and are not binding. We reserve the right to revise or upgrade the specification for improvement.

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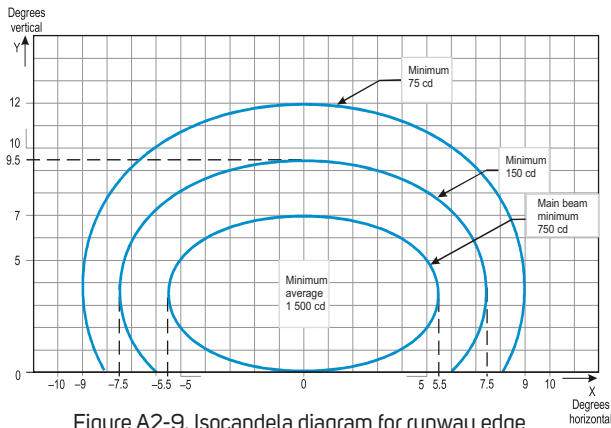


Figure A2-9. Isocandela diagram for runway edge light where width of runway is 45 m (Red light)

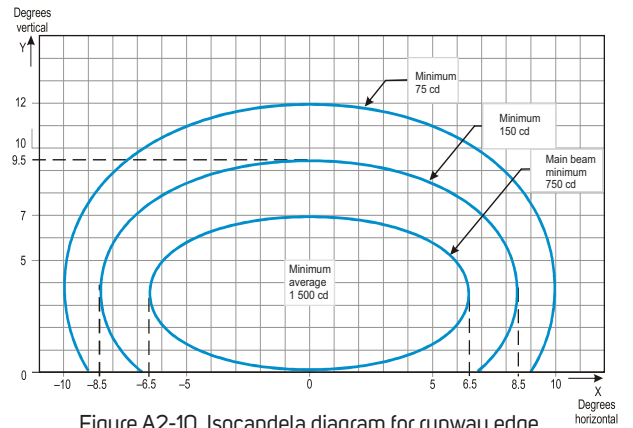
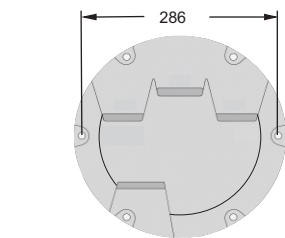


Figure A2-10. Isocandela diagram for runway edge light where width of runway is 60 m (Red light)

Runway Edge Light ICAO Fig. A2-9 / Fig. A2-10				Omni Required Min. Average Cd
Elevated Angle	3.5°	3.5°	3.5°	50cd
Toed-In Angle	3.5° / 4.5°	3.5° / 4.5°	3.5° / 4.5°	
Max./Min. Intensity Ratio	<3.0	<3.0	<3.0	
Color	White	Yellow	Red	
Main Ellipse Average Intensity	10,000cd	4,000cd	15,000cd	
Main Ellipse Minimum Intensity	5,000cd	2,000cd	750cd	
Second Ellipse Minimum Intensity	1,000cd	4,00cd	150cd	
Third Ellipse Minimum Intensity	500cd	2,00cd	75cd	
Chromaticity	ICAO Annex. 14 Fig A1-1b			

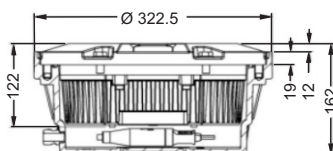
Dimensions



(Top View)

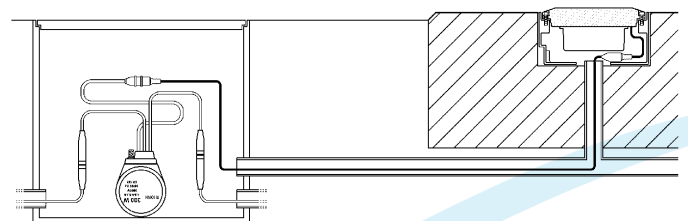
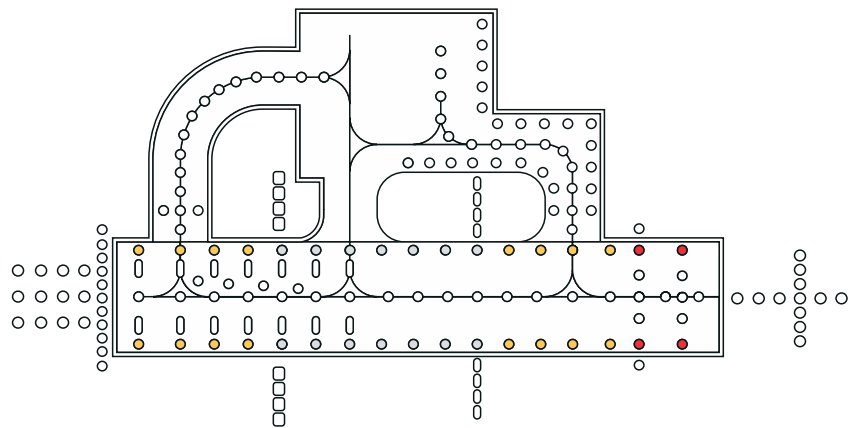


(Front View)



(Section View)

Installation



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